

TODD WILLIAM ERICKSON

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SENIOR SOFTWARE ENGINEER

As an experienced software engineer with almost 1 year as a contractor at American Express, 3 years software development at Amazon, 8 years of engineering at Boeing, 2 master's degrees in engineering, and skills with both frontend and backend services – as well as the cloud services that allow both to operate at scale – I am seeking to join a high performing technology development team to leverage my passion for software development in order to innovate and exceed customer expectations.

TECHNICAL SKILLS

Languages - Kotlin, Java, TypeScript, JavaScript, Python, C, C++, Shell scripting

Tools - Git, IntelliJ IDEA, VS Code, Maven, Gradle, Docker, Linux

Frameworks & Libraries - NestJS, React, Node.js, Bootstrap, HTML, CSS, jQuery, i18n, Lombok, Dagger2

Databases - SQL, PostgreSQL, NoSQL, MongoDB, DynamoDB

AWS Cloud - AWS CDK, Lambda, API Gateway, SQS, S3, CloudWatch

Architecture & Design - Hexagonal Architecture, CQRS, Event Sourcing, Dependency Injection

Testing - Unit Testing, Integration Testing, E2E Testing, JUnit, Kotest, MockK, Mockito, Cypress, Jest

Engineering Analysis - MATLAB, Simulink, Simscape, Easy5, ANSYS CFX, ANSYS Fluent, ANSYS ICEM CFD, SolidWorks Simulation, Rhinoceros 3D, V-Ray, SolidWorks, CATIA

PROFESSIONAL EXPERIENCE

TEKsystems | Contractor – American Express

2024 - Present

Software Development Engineer

As a part of the OneMarket team, used NestJS, Node.js, React, the OneApp framework, and OneData functions to create full-stack tools to enable Marketing and Search Engine Optimization (SEO) for all American Express resident websites.

Key Accomplishments

- Helped to guide the system architecture, including CQRS and Event driven systems
- Mentored and guided other engineers on software best practices
- Developed an SDK to enable developers to add Structured Data to their websites, enabling rich search results

Amazon | Multi-Channel Fulfillment (MCF)

2021 - 2024

Software Development Engineer

As a part of an Agile DevOps team, designed, developed, and maintained software using Java, Kotlin, Typescript, Python, Linux and AWS. Other tasks include on call and maintenance of full CI/CD pipelines.

Key Accomplishments

- Preserved 11% of MCF units by the creation of a productionalized fulfillment client, MCF Shopify app & Shopify fulfillment client adapter, created parts of the CDK infrastructure, added handling of Shopify's Mandatory GDPR webhooks, and implemented key steps in the app's installation, fulfillment creation, and order update workflows.
- Created a custom, HTTP based, generically typed GraphQL client in Kotlin. Created Shopify specific instantiation of client and templated initial calls which were used in the creation of 21 calls to Shopify.
- Designed and created a Typescript React website for the supplychain.amazon.com order management dashboard as a part of an effort to enable a total operating profit opportunity of \$400M. Created using AWS CDK, Cypress integration tests, custom react hooks, and full i18n. Designed a schema first GraphQL API as the backend interface, creating both a GraphQL schema with a self-documenting GraphiQL IDE.
- Led Amazon to avoid 860 tickets and avoid the loss of 3.5M shipping unit opportunities by enabling the intake of inventory to the correct fulfillment centers for merchants who require unbranded packaging for shipments.

Master's Degree | Computer Science (*Professional Break*)

2019 - 2020

Boeing | Commercial Airplanes Division**2011 - 2018****Propulsion Engineer**

Developed and maintained MATLAB, Simulink, and Simscape based fuel system analysis tools. Performed analysis on enterprise-wide aircraft fuel systems analysis using MATLAB and Simulink. Researched and consulted with enterprise customers on the effects of fuel properties on aircraft fuel systems.

Key Accomplishments

- Participated in Boeing's Product Development Grand Challenges Competition 4 times. Developed projects including a family of 5 single aisle aircraft with 80% parts in common, a firefighting artillery shell which won the Bold Ingenuity: Inspired Visionary Creativity award, and a 737 replacement with open-fan engines which won the Best Overall Innovation: Product Differentiation award. Together the projects led to 9 patents.
- Created MATLAB class-based ISO 10303-21 STEP file interpreter for generating Simulink tubing geometry.
- Modeled and performed certification analysis on the KC-46 aerial refueling system's surge pressures.
- Developed software for estimating the vapor pressure of fuels using MATLAB for use in aircraft certification.
- Took the initiative to learn HTML, CSS, JavaScript, and jQuery and update the Fuels Research group's website

EDUCATION**University of Southern California, Los Angeles, CA****Master of Science Computer Science, GPA 3.57****Graduated 2020**

- Implemented the processes, threads, virtual file system, and the virtual memory of the weenix kernel in C.
- Created Azure webapps using a news API and JavaScript, Flask, and Python; and React, Bootstrap, and Node.js.
- Built a full stack Android news aggregation app using Java with The Guardian and OpenWeatherMap APIs.
- Created an MLP artificial neural network to classify hand-written digits (0-9) using C++ from scratch.
- Built a Go 5x5 board game AI playing agent, implementing Monte Carlo and Minimax search using C++.
- Created a multi-process TCP and UDP socket networking system in C and C++.
- Led two other students to create a React app for the COCOMO software cost estimation tool, integrating GitLab CI/CD to perform the DevOps tasks of build with Maven, test, and deploy on an Apache server.

Master of Science Aerospace Engineering, GPA 3.40**Graduated 2011****Bachelor of Science Mechanical Engineering, Cum Laude, GPA 3.58****Graduated 2010**

- AIAA Undergraduate Team Aircraft Design Award

PATENTS

<i>System and method for augmenting a primary powerplant</i>	Pending	US 20180118364
<i>Fire-retarding artillery shell</i>	Granted	US 10955227
<i>Laterally reinforced variable pitch rotor</i>	Granted	US 10018058
<i>Systems and methods for determining sizes and shapes of geodesic modules</i>	Granted	US 9965582
<i>Systems and methods for manufacturing a tubular structure</i>	Granted	US 9957031
<i>Integrated pusher turbofan for aircraft</i>	Granted	US 9950800
<i>Contra-rotating open fan propulsion system</i>	Granted	US 9835093
<i>Vibration dampening for horizontal stabilizers</i>	Granted	US 9828084
<i>Geodesic structure forming systems and methods</i>	Granted	US 9789548